

Linear Algebra HW 8

Robert O'Connell

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Question 1

$\chi_A(\lambda)$ is of degree 11 $\Rightarrow A \in M_{11 \times 11}(\mathbb{C})$

- $\lambda = 0$, algebraic multiplicity 6, largest block associated with is 3×3
- $\lambda = 4$, algebraic multiplicity 3, largest block associated with is 2×2
- $\lambda = -5$, algebraic multiplicity 2, largest block associated with is 2×2

$\dim(\mathcal{C}(A)) = 8 \Rightarrow \dim(\mathcal{N}(A)) = 3$

$\Rightarrow 3$ linearly independent eigenvectors associated with $\lambda = 0$

$$\Rightarrow J = J_2(4) \oplus J_1(4) \oplus J_2(-5) \oplus J_3(0) \oplus J_2(0) \oplus J_1(0)$$

Question 2

To check whether the two matrices are similar, I will find the Jordan Normal Form of each of them and check if it is the same

$\mathcal{N}(A - I)$:

$$\begin{bmatrix} 0 & 1 & 0 & 1 & | & 0 \\ -2 & -1 & 4 & 1 & | & 0 \\ -1 & 0 & 2 & 1 & | & 0 \\ -2 & -3 & 4 & 3 & | & 0 \end{bmatrix}$$

Which row reduces to

$$\begin{bmatrix} 1 & 0 & -2 & -1 & | & 0 \\ 0 & 1 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Which has two free variables, thus $\dim(\mathcal{N}(A - I)) = 2$

$\mathcal{N}(A - 3I)$:

$$\begin{bmatrix} -2 & 1 & 0 & 1 & | & 0 \\ -2 & -3 & 4 & 1 & | & 0 \\ -1 & 0 & 0 & 1 & | & 0 \\ -2 & -3 & 4 & 1 & | & 0 \end{bmatrix}$$

Which row reduces to

$$\begin{bmatrix} 1 & 0 & 0 & -1 & | & 0 \\ 0 & 1 & 0 & 3 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Which has two free variables, thus $\dim(\mathcal{N}(A - 3I)) = 2$

Thus, the Jordan form of A is

$$J_A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$\mathcal{N}(B - I) :$

$$\begin{bmatrix} -2 & 2 & 2 & 0 & | & 0 \\ -2 & 2 & 2 & 0 & | & 0 \\ -2 & 2 & 2 & 0 & | & 0 \\ -2 & 0 & 4 & 2 & | & 0 \end{bmatrix}$$

Which row reduces to

$$\begin{bmatrix} -2 & 2 & 2 & 0 & | & 0 \\ 0 & -2 & 0 & 2 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Which has two free variables, thus $\dim(\mathcal{N}(B - I)) = 2$

$\mathcal{N}(B - 3I) :$

$$\begin{bmatrix} -4 & 2 & 2 & 0 & | & 0 \\ -2 & 0 & 2 & 0 & | & 0 \\ -2 & 2 & 0 & 0 & | & 0 \\ -2 & 0 & 4 & 0 & | & 0 \end{bmatrix}$$

Which row reduces to

$$\begin{bmatrix} -4 & 2 & 2 & 0 & | & 0 \\ 0 & -1 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Which has two free variables, thus $\dim(\mathcal{N}(B - 3I)) = 2$

Thus, the Jordan form of B is

$$J_B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

Since A and B have the same Jordan Normal Form, they are similar to each other

Question 3

If we use the coordinate mapping with respect to the basis $\mathcal{E} = \{x^3, x^2, x, 1\}$, this question becomes one from $\mathbb{R} \rightarrow \mathbb{R}$, which is much easier

One such linear operator is then given by the matrix

$$[T]_{\mathcal{E}, \mathcal{E}} = \begin{bmatrix} -3 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

Then converting back we get

$$T(ax^3 + bx^2 + cx + d) = -3ax^3 - 3bx^2 + 2cx + 2d$$

Question 4

$\Rightarrow$

Let A be an $n \times n$ invertible matrix, with minimal polynomial $m(x) = \sum_{i=0}^n a_i x^i$

A invertible $\iff \dim(\mathcal{N}(A)) = 0$

$\iff 0$ not an eigenvalue of A

$\iff m(0) \neq 0$

$\Rightarrow a_0 \neq 0$

$\Leftarrow$

Let A be an $n \times n$ matrix, with minimal polynomial $m(x) = \sum_{i=0}^n a_i x^i = \sum_{i=1}^n a_i x^i + a_0$, where $a_0 \neq 0$

$m(A) = 0 \iff \sum_{i=1}^n a_i A^i + a_0 I = 0$

$\iff -\frac{1}{a_0} \sum_{i=1}^n a_i A^i = I$

$\iff A(-\frac{1}{a_0} \sum_{i=1}^n a_i A^{i-1}) = I$

$\Rightarrow AA^{-1} = I$

Question 5

(a)

A is an upper triangular matrix, thus $\lambda = 1$ is a triple eigenvalue $\mathcal{N}(A - I)$:

$$\left[\begin{array}{ccc|c} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Which has two free variable, thus $\dim(\mathcal{N}(A - I)) = 2$

Thus the Jordan Normal Form of A is

$$J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b)

$\chi_A(x) = (1 - x)^3$, thus $m(x) = \pm(1 - x)^k$, for some $1 \leq k \leq 3$, we also need $m(A) = 0$

$$(I - A) = \begin{bmatrix} 0 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \neq 0$$

$$(I - A)^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus $m(x) = (1 - x)^2$

(c)

$$A^2 = \begin{bmatrix} 1 & 2 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^3 = A^2 A = \begin{bmatrix} 1 & 3 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Then

$$A^6 = (A^3)^2 = \begin{bmatrix} 1 & 6 & -6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question 6

(a)

$\mathcal{N}(A - I)$:

$$\left[\begin{array}{ccc|c} -50 & 43 & 7 & 0 \\ -57 & 49 & 8 & 0 \\ -8 & 7 & 1 & 0 \end{array} \right]$$

Which row reduces to

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Which has one free variable, thus $\dim(\mathcal{N}(A - I)) = 1$

Then A has the Jordan Form

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

(b)

$\chi_A(x) = (1 - x)^3$, since a matrix and its Jordan form will have the same characteristic polynomial thus $m(x) = \pm(1 - x)^k$, for some $1 \leq k \leq 3$, we also need $m(A) = 0$

$$(I - A) = \begin{bmatrix} 50 & -43 & -7 \\ 57 & -49 & -8 \\ 8 & -7 & -1 \end{bmatrix} \neq 0$$

$$(I - A)^2 = \begin{bmatrix} -7 & 6 & 1 \\ -7 & 6 & 1 \\ -7 & 6 & 1 \end{bmatrix} \neq 0$$

$$(I - A)^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus $m(x) = (-1)(1 - x)^3$

(c)

We know $m(A) = 0$, thus

$$\Rightarrow (I - A)^3 = 0$$

$$\Rightarrow I - 3A + 3A^2 - A^3 = 0$$

$$\Rightarrow A(A^2 - 3A + 3I) = I$$

$$\Rightarrow A^{-1} = A^2 - 3A + 3I$$

$$\Rightarrow A^{-1} = \begin{bmatrix} -106 & 92 & 15 \\ -121 & 105 & 17 \\ -23 & 20 & 4 \end{bmatrix} + \begin{bmatrix} 147 & -129 & -21 \\ 171 & -150 & -24 \\ 24 & -21 & -6 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 142 & -37 & -6 \\ 50 & -42 & -7 \\ 1 & -1 & 1 \end{bmatrix}$$